Praxis Precision Medicines Short Recommendation

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Executive Summary

Praxis Precision Medicines (NASDAQ:PRAX), a clinical-stage biopharma company, is anticipating a major Q1 2025 catalyst: an interim phase 3 readout on leading asset ulixacaltamide (PRAX-994) for essential tremor (ET). Due to a string of failures in similar compounds for ET from other biopharma companies as well as weak Phase 2 results, this report argues the readout will fail to show efficacy, leading to ulixacaltamide's discontinuation for ET and a drop in share price. I assign an 85% probability to this outcome and predict a share price decline of 65-75% from $80/share to $20-28/share upon this catalyst.

Background

Essential tremor (ET) is a neurological disorder causing involuntary, rhythmic shaking, primarily in the hands but also affecting the head, voice, and other areas. As of 2023, it affects about 7 million people in the U.S., making it a leading movement disorder.[1] Current first-line treatments, propranolol (a beta-blocker) and primidone (an anticonvulsant), can reduce symptoms, but 30–50% of patients do not respond, highlighting the need for new therapies.[2] No treatments are specifically approved for ET.

PRAX-994, a potential alternative or adjunctive therapy, targets the cerebello-thalamo-cortical (CTC) circuit thought to drive ET via a selective T-type calcium channel blocker (TTCCB). Similar TTCCBs suvecaltamide (Jazz Pharmaceuticals) and NBI-827104 (Neurocrine Bioscience), have been discontinued for ET due to lack of efficacy in Phase 2 trials.

ET's key clinical endpoint is The Essential Tremor Rating Assessment Scale (TETRAS) consisting of two subscales: the activities of daily living (ADL) subscale, which assesses the subjective impact of ET on daily tasks, and the performance subscale (PS), which evaluates functional impacts like spiral drawing and handwriting.[3] The FDA's recommended endpoint for ET trials includes 11 ADL items (excluding social life) and two PS items (spiral drawing and handwriting).[4] This modified endpoint, scored on a 0–3 Likert scale, is called the "mADL."

Review of the Clinical Evidence

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[1] Louis ED, Ottman R. How many people in the USA have essential tremor? Deriving a population estimate based on epidemiological data. *Tremor Other Hyperkinet Mov (N Y)*. 2014;4:259. Published 2014 Aug 14. doi:10.7916/D8TT4P4B

[2] Koller WC, Vetere-Overfield B. Acute and chronic effects of propranolol and primidone in essential tremor. *Neurology*. 1989;39(12):1587-1588. doi:10.1212/wnl.39.12.1587

[3] Elble RJ. The Essential Tremor Rating Assessment Scale. J Neurol Neuromed. 2016;1(4):34-38.

[4] https://www.sec.gov/Archives/edgar/data/1689548/000168954823000056/form8-kxexhibit9922.htm

Neurocrine's data

Neurocrine's TTCCB, NBI-827104, was tested in a double-blind, placebo-controlled, cross-over phase 2 trial with 31 patients (NCT04880616).[5] The study measured changes from baseline in TETRAS PS and ADL after 28 days for each period (placebo and NBI-827104, separated by a 14-day washout). Endpoints were calculated as the average difference in each metric between the two periods per patient. No significant differences were found across any endpoints ($p \geq 0.05$), including TETRAS and ADL, leading to the therapy's discontinuation.

Some may argue that the trial is underpowered with only 31 patients. However, the cross-over design efficiently controls for inter-patient variability, including rater and subject tendencies and baseline factors like medication use. Roughly speaking, the power is equivalent to a parallel design of 62 patients or possibly even more, depending on the degree of variance reduction. Another critique might be the 28-day duration, but, as Jazz's data will show, this timeframe can be sufficient to detect effects.

Jazz's data

Jazz's Phase 2a T-CALM study (NCT03101241) enrolled 92 patients in a multicenter, double-blind, placebo-controlled, parallel design trial.[6] The primary endpoint, the change from baseline in TETRAS-PS scored by independent video raters after 28 days, was not statistically significant (least squares [LS] mean: -1.8 for CX-8998 vs. -2.3 for placebo; $p = 0.696$). Nevertheless, secondary endpoints showed significant improvements in symptom reduction in investigator-rated TETRAS-PS (-4.8 for CX-8998 vs. -2.8 for placebo; $p = 0.017$) and TETRAS-ADL (-4.5 vs. -1.6; $p = 0.049$). The LS means reflect adjustments for anti-tremor medication use, study site, and baseline TETRAS-PS via ANCOVA.

There are three key observations from the T-CALM results. First, the p-values are only nominally significant at the 0.05 level, with no adjustment for multiple testing. A simple Bonferroni correction for the six tests presented in the manuscript would render all p-values insignificant. Even so, a potential signal cannot be entirely ruled out. Second, there is concordance between TETRAS-ADL and TETRAS-PS, as CX-8998 nominally outperformed placebo for both endpoints. Third, the disagreement between video raters and investigators for TETRAS-PS suggests that results depend on the evaluation method, with inter-rater reliability notably lower for investigators (ICC = 0.6) compared to video raters (ICC = 0.8).

Despite the tepid results of T-CALM, Jazz conducted a second, larger trial (NCT05122650) with 420 patients, indicating it was powered a priori. The primary endpoint shifted to measure the difference in mADL after 12 weeks between CX-8998 (renamed

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[5] C. de Cuba, L. Smits, P. Kremer, R-J. Doll, J. Heuberger, R. Helmich, A F. van Rootselaar, R. Elble, W. Ondo, L. Rees, N. Wedick, G. Liang, D. Haubenberger. An evaluation of NBI-827104 for essential tremor: results from a randomized, double-blind, placebo-controlled, dose-escalation, crossover study [abstract]. *Mov Disord*. 2023; 38 (suppl 1).

[6] Papapetropoulos S, Lee MS, Versavel S, et al. A Phase 2 Proof-of-Concept, Randomized, Placebo-Controlled Trial of CX-8998 in Essential Tremor. *Mov Disord*. 2021;36(8):1944-1949. doi:10.1002/mds.28584

JZP385) and placebo. Jazz announced on June 20, 2024 that the trial failed to meet its primary or exploratory endpoints and that they were discontinuing their ET program.[7] As of December 2024, no other data of this trial has been reported.

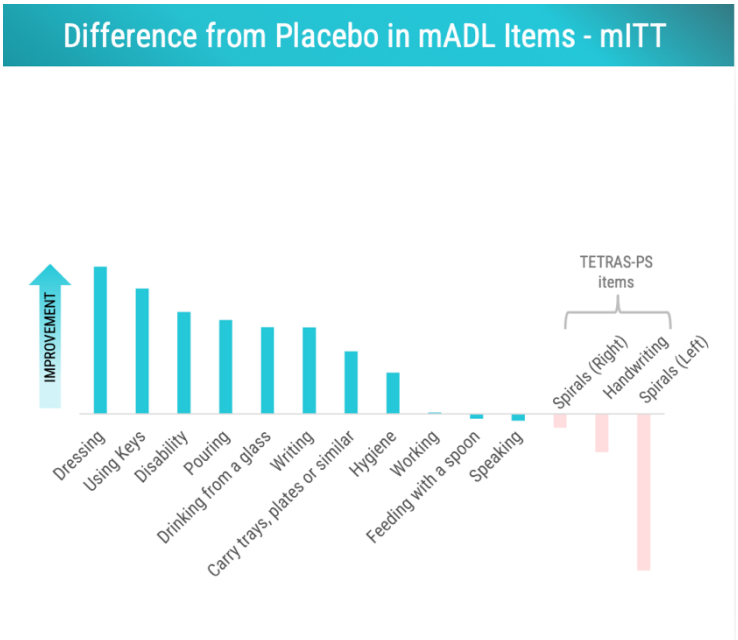
It is important to note that extending the trial to 12 weeks and increasing the sample size to 420 patients did not yield meaningful separation between treatment and placebo, reinforcing the argument for TTCCB inefficacy for ET. These results, the most robust data to date, were reported after PRAX initiated its pivotal Phase 3 trial in November 2023—the key catalyst discussed in this report.

PRAX's ESSENTIAL1

In the ESSENTIAL1 study (NCT05021991), PRAX-994 was evaluated in a multicenter Phase 2 trial where 116 patients were randomized 2:1 to PRAX-994 or placebo, with the primary endpoint being the change in mADL after 56 days as assessed by investigators.[8]This primary endpoint did not achieve statistical significance, with an LS mean reduction of -3.01 for PRAX-994 and -1.44 for placebo ($p = 0.126$), adjusted for baseline value, propranolol use, and familial history of ET. A post-hoc analysis, excluding TETRAS-PS items and focusing solely on ADL questions 1-11, revealed a nominally significant difference, with reductions of -2.61 for PRAX-994 versus -0.88 for placebo ($p = 0.042$).

I contend that the post-hoc result is spurious (as is the case with most post-hoc analyses in general). Beyond the p-value being barely nominally significant at 0.05, there is no information on the number of post-hoc tests conducted, which is a major issue in attempting to guess an appropriate multiplicity adjustment. For instance, PRAX could have tested removing each PS question individually but still failed to find significance.

Moreover, the post-hoc result lacks mechanistic consistency. As shown in the figure, PRAX-994 performed well on ADL questions but poorly on PS questions. For example, patient-reported "Writing" improved, yet objective measures like "Handwriting" and "Spirals" showed worse outcomes than placebo. If a drug truly reduced ET symptoms, one would expect consistent improvements across both subjective and objective outcomes. While Jazz's T-CALM study demonstrated alignment



[7] https://investor.jazzpharma.com/news-releases/news-release-details/jazz-pharmaceuticals-provides-update-phase-2b-trial

[8] https://ir.praxismedicines.com/static-files/625750d7-3062-48e2-a29c-4b42af61123d

between ADL and PS improvements compared to placebo, ESSENTIAL1 instead showed discordance. This suggests spurious findings at best and inefficacy at worst.

The Pivotal Phase 3 ESSENTIAL3 Trial

Thus far, I have highlighted that both Neurocrine and Jazz have discontinued their ET programs due to failure to show efficacy and ESSENTIAL1 was essentially a negative trial with a potentially spurious post-hoc result. This history suggests a low probability of success for ESSENTIAL3, the pivotal Phase 3 trial.

ESSENTIAL3 randomized 400 patients 1:1 to PRAX-994 or placebo with a modified primary endpoint of mADL1-11 over 12 weeks (NCT06087276), likely influenced by ESSENTIAL1's post-hoc analysis. However, FDA communications indicate a preference for the original mADL endpoint over ESSENTIAL3's modification.[4] While inclusion-exclusion criteria remain largely unchanged from ESSENTIAL1 and other trials, investigator raters were replaced with telehealth and in-home visits, which, as seen in T-CALM, can affect performance results.

Predicting ESSENTIAL3 outcomes based on Jazz's larger Phase 2b trial is reasonable: both had similar sample sizes and a 12-week duration. Even though the primary endpoint may be different, Jazz's Phase 2b tested several secondary endpoints including the ADL scale. Given that Jazz decided to discontinue its ET program, it would be reasonable to assume there were no meaningful results in any of these secondary endpoints.

Scenarios and Probabilities (current share price as of December 1st, 2024)

1. The interim analysis of ESSENTIAL3 due Q1 2025 will not meet statistical significance and PRAX's ET program will be discontinued. I assign an 85% probability to this outcome. On the day PRAX presented the results in ESSENTIAL1, their stock declined 65% to $14.95. Since there are other assets in the pipeline due for Phase 2 results soon after this catalyst, I predict a similar drop of 65-75% in the share price from roughly $80/share to $20-28/share.
2. The interim analysis of ESSENTIAL3 will not meet statistical significance nor the futility bound but based on a small numerical separation, management will decide to continue the trial. I assign a 10% probability to this outcome. The stock will decline but at a lesser magnitude of 50-60% in the share price from roughly $80/share to $32-45/share.
3. Given the trial is allowed to continue to its end on a non-significant result (15% probability), there is a 10% probability it meets ESSENTIAL3 primary endpoint. Multiplying the probabilities gives us a 1.5% probability of positive results. Even if the results are positive, there is likely to be regulatory pushback as the FDA will request a re-analysis of the data to include the TETRAS PS items, and doing so will likely yield results that fail to demonstrate efficacy. At this point, there will be data from other programs, so it is not possible to predict the decline in the share price independent from these other data.

4. The interim analysis demonstrates a statistically significant positive result on mADL1-11 (probability 5%). The stock will increase 70-80% from $80/share to $136-144/share. Nevertheless, as I have highlighted, the FDA may issue a CRL asking for a re-analysis on their preferred endpoint, which could be insignificant. This would present another short opportunity.